

TOPEKA FORBES, KS, USA

WMO: 724565

Lat: 38.950N

Lon: 95.664W

Elev: 1067

StdP: 14.14

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13920

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.6	8.8	-5.2	4.3	5.9	-0.8	5.5	10.5	28.6	38.5	26.4	34.5	9.3	320	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	98.9	76.4	95.0	76.1	92.0	75.2	79.3	92.4	78.1	90.8	76.9	88.9	12.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	138.8	86.2	74.4	133.6	85.1	73.1	127.9	83.7	43.5	92.5	42.3	90.8	41.0	88.9	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.1	23.3	20.2	-4.4	102.9	5.1	3.7	-8.1	105.5	-11.1	107.6	-13.9	109.7	-17.7	112.4	
			DB												
			WB	-5.1	81.9	4.9	1.7	-8.6	83.1	-11.5	84.1	-14.2	85.1	-17.8	86.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	30.1	33.8	45.5	55.7	65.6	75.0	79.6	77.9	69.9	56.8	44.8	33.4
	DBStd	19.62	11.26	11.82	11.13	9.43	8.09	6.46	5.97	6.05	7.95	9.54	10.15	10.88
	HDD50	2135	618	468	219	49	3	0	0	0	42	214	522	
	HDD65	4881	1081	874	609	303	95	7	0	1	40	285	607	979
	CDD50	4247	2	14	80	219	486	749	918	864	596	254	58	7
	CDD65	1519	0	0	5	23	113	306	453	400	186	32	1	0
	CDH74	16227	0	3	56	262	1083	3185	5228	4271	1818	306	15	0
	CDH80	7304	0	0	7	52	335	1354	2652	2095	729	79	1	0
Wind	WSAvg	9.4	10.0	10.3	10.9	11.7	9.5	9.1	7.6	7.2	7.8	9.1	9.9	9.9
Precipitation	PrecAvg	39.2	0.9	1.6	2.2	4.0	5.3	5.4	4.3	4.5	3.8	3.1	2.2	1.4
	PrecMax	52.6	2.7	4.1	4.9	8.8	11.7	8.5	13.9	8.1	8.3	7.3	8.6	3.7
	PrecMin	27.6	0.0	0.0	0.3	1.0	1.6	1.9	1.2	1.1	0.8	0.2	0.1	0.1
	PrecStd	7.1	0.7	1.0	1.3	2.1	2.6	1.9	3.0	2.1	2.1	1.7	2.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.2	72.1	79.9	85.2	90.9	97.5	102.7	102.4	97.3	87.3	76.1	66.0
		MCWB	50.2	55.1	62.7	66.0	72.4	76.8	75.1	75.1	73.8	69.6	61.2	57.6
	2%	DB	57.3	64.0	74.1	80.9	87.3	93.6	99.3	98.6	91.3	81.6	71.0	60.1
		MCWB	46.3	52.7	58.7	64.5	70.5	75.7	77.2	75.8	73.6	65.6	58.2	50.9
	5%	DB	52.0	58.3	69.7	76.6	83.8	90.8	95.8	94.1	88.2	76.8	65.8	54.5
		MCWB	43.6	48.8	57.0	61.8	69.1	75.1	76.9	75.4	72.6	63.9	55.5	47.6
	10%	DB	46.7	52.8	64.2	72.5	80.6	87.9	92.3	90.7	83.9	72.7	61.3	49.9
		MCWB	40.1	44.8	54.1	60.4	68.2	73.8	76.1	74.7	70.4	62.1	53.0	43.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	59.1	65.9	69.3	76.0	80.3	81.4	80.3	77.8	73.1	63.9	60.1
		MCDB	58.7	65.4	76.5	81.2	86.7	91.6	94.6	92.7	90.4	82.5	71.3	64.5
	2%	WB	48.0	53.9	62.3	66.9	73.7	78.0	79.6	78.6	75.6	70.0	60.2	53.7
		MCDB	55.4	61.4	69.4	76.5	83.5	89.8	93.4	91.5	86.7	77.3	67.4	57.5
	5%	WB	44.3	49.6	59.1	64.4	71.4	76.5	78.4	77.4	74.2	66.7	57.4	48.1
		MCDB	51.4	58.5	66.7	73.3	80.4	87.9	91.5	89.9	84.9	73.5	64.2	53.7
	10%	WB	40.2	44.4	54.7	61.9	69.4	75.0	77.3	75.9	72.4	63.4	53.8	43.8
		MCDB	47.1	52.2	63.7	71.1	77.9	85.4	89.7	87.3	82.1	69.9	60.3	49.8
Mean Daily Temperature Range	5% DB	MDBR	20.2	20.9	22.5	22.5	21.3	20.4	20.9	21.5	22.4	22.4	21.8	19.5
		MCDBR	29.2	30.3	30.1	28.2	25.0	22.7	24.7	25.7	24.9	26.6	28.4	26.9
		MCWBR	19.7	20.1	17.3	15.8	11.9	9.5	8.4	8.9	10.2	13.9	17.2	19.0
	5% WB	MCDBR	27.3	27.6	23.2	23.4	21.3	20.8	21.6	22.3	21.6	20.7	23.4	24.2
	MCWBR	19.3	19.6	14.7	15.1	11.9	10.2	9.1	9.4	10.2	13.2	16.1	19.0	
Clear-Sky Solar Irradiance	taub	0.287	0.301	0.330	0.381	0.414	0.428	0.430	0.431	0.383	0.339	0.305	0.283	
	taud	2.485	2.457	2.428	2.308	2.275	2.264	2.286	2.287	2.371	2.481	2.513	2.513	
	Ebn at Noon	285	295	295	284	275	271	269	266	273	277	275	278	
	Edh at Noon	25	29	33	40	42	43	42	41	35	28	24	23	
All-Sky Solar Radiation	RadAvg	722	972	1259	1601	1831	2076	2059	1838	1525	1108	790	620	
	RadStd	70	109	102	104	159	139	138	94	105	125	71	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page